Weekly Literature Review

Period: Nov 04 – Nov 11, 2026

This Week in Brief

The biggest news this week is that two flagship novel-target outcome trials failed despite deep biomarker lowering: pelacarsen for Lp(a) and ziltivekimab for IL-6. Both results undermine these targets and cast doubt on IL-6 programs in HFpEF. Strategy trials fared better, with left bundle-branch pacing beating RV pacing in AV block and a systolic target below 120 mm Hg reducing events after stroke, though a diabetes trial was neutral. For transplant practice, the first randomized trial of dapagliflozin in heart transplant recipients showed a modest eGFR-slope benefit without excess infections.

Cardiology

A heavy trial week brought a split verdict. Two flagship novel-target outcome trials failed despite large biomarker effects: pelacarsen for Lp(a) and ziltivekimab for IL-6. Biomarker lowering again did not reliably predict clinical benefit. Strategy and device trials did better. Conduction system pacing beat RV pacing, and a systolic target below 120 mm Hg reduced events after stroke. SGLT2 inhibition also showed a modest kidney benefit in heart transplant recipients.

Pelacarsen and Cardiovascular Outcomes in Patients with High Lipoprotein(a)

N Engl J Med · Cho L, Nicholls SJ, Frndak S, et al. · Placebo-controlled RCT, 8,350 patients · DOI

Area: Lipoprotein(a) lowering, secondary prevention

Design: Patients with established cardiovascular disease and Lp(a) of at least 70 mg/dL were randomized to monthly subcutaneous pelacarsen 80 mg or placebo. The primary endpoint was cardiovascular death, nonfatal MI, nonfatal stroke, or urgent coronary revascularization. Median follow-up was 4.7 years.

Results: Median baseline Lp(a) was 236 nmol/L. At 12 months, Lp(a) was 64.4 nmol/L with pelacarsen vs 233.4 nmol/L with placebo. The primary endpoint occurred in 12.3% vs 12.1% (HR 1.02, 95% CI 0.90 to 1.15; P = 0.76). In patients with Lp(a) of at least 90 mg/dL, rates were 12.3% vs 12.3% (HR 1.00; 95% CI 0.87 to 1.15). Serious adverse events occurred in 37.3% vs 41.1%.

Context: The first Lp(a) outcomes trial was flatly negative despite deep Lp(a) lowering, with no signal even in the highest-Lp(a) stratum. Mendelian randomization predicted that very large absolute reductions would be needed for benefit, and on-treatment levels near 64.4 nmol/L may still sit above any protective threshold. This is a serious blow to Lp(a) as a modifiable target. Attention now shifts to the siRNA outcome trials, which achieve deeper suppression. Until those report, measuring Lp(a) remains a risk-stratification tool, not a treatment target.

Interleukin-6 Inhibition with Ziltivekimab for Atherosclerotic Disease

N Engl J Med · Ridker PM, Lincoff AM, Marx N, et al. · Double-blind RCT, 6,376 patients · DOI

Area: Anti-inflammatory therapy, ASCVD with CKD

Design: Adults with atherosclerotic disease, chronic kidney disease, and hsCRP of at least 2 mg/L were randomized to subcutaneous ziltivekimab 15 mg monthly or placebo. The primary endpoint was 3-point MACE. Median follow-up was 33 months.

Results: At 24 months, the between-group difference in median percent change was -76 percentage points for free IL-6 and -64 percentage points for hsCRP. The primary endpoint occurred in 17.6% with ziltivekimab vs 17.7% with placebo (HR 0.99; 95% CI 0.88 to 1.11; P = 0.84). Serious infections occurred in 20.4% vs 17.9%; overall serious adverse events were 54.6% vs 54.8%.

Context: Potent IL-6 suppression produced zero MACE benefit. The population, CKD with elevated hsCRP, was the one genetic and observational data most favored for this target. The residual-inflammatory-risk paradigm now rests mainly on CANTOS and the colchicine trials, and those results have themselves been inconsistent, with a neutral post-MI colchicine trial. The infection excess without benefit likely ends IL-6 blockade for atherosclerosis. It also suggests hsCRP is a risk marker more than a treatment target. Ongoing IL-6 programs in HFpEF now carry much more uncertainty.

Left Bundle-Branch Pacing versus Right Ventricular Pacing for Atrioventricular Block

N Engl J Med · Qian Z, Jiang L, Hou X, et al. · Open-label RCT, 702 patients · DOI

Area: Conduction system pacing, AV block

Design: Patients with AV block were randomized to left bundle-branch pacing or RV pacing. The primary endpoint was a composite of all-cause death, HF hospitalization, or pacing-induced cardiomyopathy. Median follow-up was 35.1 months, and the trial was stopped early at a prespecified interim analysis.

Results: In the modified ITT population (685 patients), primary events occurred in 16 vs 35 patients. Four-year cumulative incidence was 8.5% vs 19.2% (HR 0.41; 95% CI 0.23 to 0.75; P = 0.003). Serious adverse events unrelated to implantation occurred in 11.8% vs 9.3% (P = 0.32).

Context: This is the first randomized outcomes evidence that left bundle-branch pacing beats RV pacing in AV block, more than halving the composite. The effect is probably overstated. The trial was open-label and stopped early, and the composite likely leans on pacing-induced cardiomyopathy, a softer, imaging-dependent component that the abstract does not break down. Even so, it will accelerate conduction system pacing as the default for patients expected to pace heavily. Long-term lead performance, extractability, and results outside expert centers remain open questions.

Intensive Blood-Pressure Control in Patients with Previous Ischemic Stroke

N Engl J Med · Sampaio Silva G, Machline-Carrion MJ, Santo K, et al. · Open-label RCT, 4,368 patients · DOI

Area: Hypertension, secondary stroke prevention

Design: Brazilian patients with elevated BP and prior ischemic stroke or TIA were randomized to a systolic target below 120 vs below 140 mm Hg. The primary composite was CV death, nonfatal MI, nonfatal stroke, or hospitalization for unstable angina or HF. Median follow-up was 3.9 years.

Results: Mean achieved SBP was 122.0 vs 135.2 mm Hg. The primary endpoint occurred in 10.9% vs 13.5% (3.25 vs 4.11 per 100 person-years; HR 0.79; 95% CI 0.67 to 0.94; P = 0.007). The composite renal outcome was more common with intensive treatment. Serious adverse events were similar.

Context: Targeting systolic pressure below 120 mm Hg after stroke or TIA cut major cardiovascular events by about a fifth. That fills the gap SPRINT left by excluding stroke patients. The companion OPTIMAL-DIABETES trial used the same targets and design but was neutral. Diabetes therefore remains the contested population, with conflicting trials on either side. The excess renal events, plus open-label measurement, argue for monitoring kidney function. Still, this supports a lower target in secondary stroke prevention than most guidelines currently specify.

Dapagliflozin for Renal Protection in Heart Transplant Recipients: A Randomized Clinical Trial

Circulation · Holmen S, Manintveld O, Damman K, et al. · Double-blind RCT, 430 patients · DOI

Area: Heart transplantation, kidney protection

Design: Heart transplant recipients at least one year post-transplant with eGFR of 25 or higher were randomized to dapagliflozin 10 mg or placebo at six Northern European centers. The primary endpoint was eGFR slope from 2 weeks to the end of 12 months of treatment.

Results: Baseline eGFR was 67, and median time from transplant was 7 years. eGFR changed by 2.0 vs 0.3 mL/min/1.73 m2, a between-group difference of 1.7 (95% CI 0.1 - 3.3; p = 0.03). There were no unexpected safety signals, and adverse event rates, including infections, were similar.

Context: This first randomized trial of an SGLT2 inhibitor in heart transplant recipients showed a small chronic eGFR-slope benefit, with no infection penalty in immunosuppressed patients. The effect is modest and surrogate-only at 12 months. The placebo arm's eGFR was essentially stable, so whether this translates into less dialysis or kidney transplantation is unproven. Combined with the broad CKD trial evidence, it reasonably supports routine dapagliflozin in stable recipients without contraindications. Hard-outcome and longer-term follow-up are needed before calling it nephroprotection against calcineurin inhibitors.

Also notable

- Long-Term Survival After Initial Invasive or Conservative Management of Stable Coronary Disease in ISCHEMIA-EXTEND (*JAMA*). ISCHEMIA-EXTEND: at 11 years, all-cause mortality was 25.8% invasive vs 26.9% conservative (adjusted HR 0.97) in stable coronary disease with moderate-severe ischemia.
- Discontinuing Statins in Older Adults without Cardiovascular Disease (*N Engl J Med*). STREAM: stopping primary-prevention statins in adults 70 or older was noninferior at 2 years, though events were 12.7% vs 9.2%.
- Hydralazine-isosorbide dinitrate in chronic heart failure with reduced ejection fraction: the H-HeFT trial (*Eur Heart J*). H-HeFT: in 592 contemporarily treated HFrEF patients, hydralazine-isosorbide dinitrate did not reduce HF events or death (HR 0.90; 95% CI 0.67-1.21).

AI in Medicine

AI is finally being tested in randomized and prospective designs, and the results are humbling. AI-ECG-guided rhythm monitoring produced only a statistically fragile gain. A year of access to a generative AI chatbot did nothing for maternal mental health. A cuffless smartwatch algorithm screened for hypertension about as well as an office reading. The tools work technically; proving that they help patients is the remaining work.

Leveraging AI-ECG Technology for Early Notification and Tracking of AF Development (LATENT): A Randomized Controlled Trial

Circulation · Lee CC, Lin WY, Liu WT, et al. · Open-label single-center RCT, 14,665 adults · DOI

Area: AI-ECG for atrial fibrillation detection

Design: Adults 40 to 85 years without prior AF/AFl were randomized to an AI-ECG-guided strategy (high-risk sinus-rhythm ECG prompts ≤7-day Holter) or usual care. Matched controls received identical monitoring. Primary endpoint: new AF, AFl, or pre-AF atrial findings within 90 days.

Results: Only 84 (1.1%) intervention and 78 (1.1%) control participants were AI-high-risk, and 48 (57.1%) high-risk intervention participants were monitored. The primary outcome occurred in 0.74% vs 0.48% (OR 1.54 [95% CI, 1.01-2.37]; P=0.046); reclassifying 1 event moved P to 0.057. Among high-risk participants, yield was 32.1% vs 2.6%. AF/AFl detection (0.3% vs 0.2%) and antiarrhythmic initiation at 180 days (2.3% vs 2.4%) did not differ.

Context: AI-ECG picked out a small group in whom Holter monitoring often found atrial abnormalities, but the population-level result hinged on a single event. The composite was carried by premature atrial ectopy rather than AF, and treatment did not change. That mirrors the broader AF screening literature: detecting more arrhythmia has not reliably translated into outcomes, and pre-AF ectopy has no established therapy. Treat this as proof that AI enrichment raises monitoring yield, not as a reason to deploy AI-ECG-triggered Holter outside trials.

Opportunistically Detecting Signs of Hypertension on a Consumer Smartwatch

Nature · Di Achille P, Cai L, Wu J, et al. · Prospective multicenter validation, 196 participants · DOI

Area: Cuffless smartwatch hypertension screening

Design: Adults without known cardiovascular disease or antihypertensive therapy wore a consumer smartwatch for seven days. An uncalibrated AI model using photoplethysmography and accelerometry was compared against 24-hour ambulatory BP monitoring as the reference, with initial office BP screening as the comparator.

Results: AI-PPG-ACC-HTN detected hypertension with sensitivity 65.8% (95% CI, 54.0%-76.3%), specificity 90.0% (95% CI, 83.2%-94.7%), and PPV 80.6% (95% CI, 68.6%-89.6%). Office BP screening had sensitivity 55.3% (95% CI, 43.4%-66.7%), the same specificity of 90.0%, and PPV 77.8% (95% CI, 64.4%-88.0%).

Context: A passive, cuffless watch algorithm matched an office reading for flagging ambulatory-confirmed hypertension, though it still missed a substantial share of cases. Framing it as a screening prompt rather than a BP measurement sidesteps the calibration and drift problems that have dogged cuffless devices. The sample is small, treated patients were excluded, and the confidence intervals overlap office screening, so superiority is not shown. Population-scale yield, false reassurance, and downstream cost remain untested.

Generative Artificial Intelligence and Maternal Health Outcomes in Pregnant and Postpartum Individuals: A Randomized Clinical Trial

JAMA Netw Open · Norell EM, Yue X, Mane H, et al. · RCT, 400 participants · DOI

Area: Patient-facing generative AI chatbot in maternal health

Design: Pregnant or postpartum (infant <6 months) individuals from racial/ethnic minoritized groups were recruited online across 49 states and DC. They were randomized 1:1 to a generative AI maternal-health chatbot or an active control (monthly children's books). Primary endpoints were GAD-7 and PHQ-9 at 12 months.

Results: The chatbot did not change anxiety (adjusted mean difference 0.10; 95% CI, -0.77 to 0.96) or depression (-0.23; 95% CI, -1.05 to 0.59). ED visits, postpartum care, contraception use, and sleep duration also did not differ. Among those pregnant at enrollment, chatbot access was associated with more self-reported pregnancy-related complications at 12 months, but not in longitudinal analyses.

Context: A year of access to a generative AI chatbot did nothing for anxiety, depression, or care use in a high-need maternal population. This is one of few randomized tests of a patient-facing LLM with long follow-up, and it undercuts the assumption that access to conversational AI translates into health benefit. The trial tested access rather than enforced engagement, and outcomes were self-reported. The complications signal is likely noise but worth watching. Patient-facing LLM tools need outcome trials, not just usability data.

AI-Identified Reasons for Nonescalation of Lipid-Lowering Therapy for Secondary Prevention

JAMA Cardiol · Retrospective cohort, 13,810 ASCVD patients · LLM chart review of lipid-lowering therapeutic inertia

An LLM extracted management reasoning from notes (F1 0.80 to 0.98 vs annotators). Lipid-lowering therapy was not escalated in 11 842 (88.7%) above-goal notes, most often citing patient-reported adverse effects (1683 [12.6%]). Single integrated system; findings reflect documentation, not actual reasoning.

A gait foundation model for prediction of multisystem health phenotypes from 3D skeletal motion

Nat Med · Foundation model, cross-sectional cohort, 3,328 adults · Gait as a multisystem biomarker (longevity)

Depth-camera 3D skeletal motion embeddings predicted age (r = 0.69) and BMI (r = 0.91), and significantly predicted 1,887 of 3,210 phenotypes beyond age, BMI, visceral fat and height. Single cohort, cross-sectional; no hard outcomes or external validation.

AI in Medical Education

This week's controlled work measures AI assistance rather than learning. AI feedback improved trainee and clinician accuracy on Doppler interpretation, but novices more often accepted wrong advice. An AI simulated-patient chatbot did not beat conventional communication training over a year. Most U.S. internal medicine clerkships still have no known AI policy and no secure tools.

AI feedback for clinician assessment of umbilical and middle cerebral artery Doppler ultrasound: a multicenter randomized trial

Am J Obstet Gynecol MFM · Bashir Z, Wong CK, Lin M, et al. · Multicenter RCT, 96 clinicians · DOI

Area: AI feedback in obstetric Doppler assessment

Design: Obstetrics and gynecology residents and specialists at five Danish hospitals were randomized 1:1 to AI visual and text feedback or none. Each judged 40 static umbilical and middle cerebral artery Doppler vignettes for gate placement and waveform adequacy. The outcome was accuracy against expert consensus; there was no real-time scanning.

Results: Accuracy was 75.0% with AI feedback vs 62.3% without (mean difference 12.7%; 95% CI 8.2-17.1; $p < 0.001$). Gains were similar in novices (70.0% vs 57.9%; 95% CI 2.4-21.7) and experienced clinicians (78.0% vs 65.0%; 95% CI 5.5-20.5). Novices followed incorrect AI feedback more often (22.3% vs 15.8%; P = .01). AI alone outperformed unassisted novices, but experienced clinicians with feedback outperformed AI alone.

Context: AI feedback raised accuracy by roughly 13 points at every experience level, but the least experienced were the most likely to adopt wrong advice. That is the central tension of AI in training: the learners who gain most are also least able to catch AI errors. The finding fits earlier work showing readers misled by erroneous AI in imaging. The trial measures assisted performance on static images, not skill retained after AI is removed, so it says nothing yet about learning or deskilling.

Policy, ethical, and practical considerations of artificial intelligence use in the internal medicine clerkship: results of a survey of clerkship directors

Acad Med · National survey, 114 of 141 U.S. schools · AI policy in internal medicine clerkships

In a national survey of clerkship directors (80.9% response), 39.5% reported no AI policy and 34.2% did not know if one existed. 73.7% would encourage AI for some clinical skill, yet only 6.1% had firewall-protected tools. Findings are self-reported and from 2024.

Evaluating an AI-Powered Patient Simulator for Communication Skills Training: One-Year Randomized Controlled Trial

J Med Internet Res · Nonblinded RCT, 50 medical students · Generative AI patient simulator for communication skills

Over one academic year, UK students practicing with the SimPatient chatbot showed a descriptively larger, but not statistically significant, gain in OSCE communication scores than conventionally trained controls. Motivation and behavioral engagement predicted improvement. The trial was small, single-site and underpowered.

Longevity Medicine

A thin week for human geroscience: no interventional trial reported outcomes. The useful human data were deflationary. A large transplant registry found no survival signal for chronic rapamycin, and a cohort found that whole-blood NAD falls with cardiovascular disease rather than with age. A new senolytic mechanism in mice is the main discovery-stage advance.

Rapamycin and human longevity among kidney transplant recipients surviving the first year

Geroscience · Registry cohort, 11,474 kidney recipients · Rapamycin and human longevity

In US kidney recipients alive with a functioning graft at 1 year, propensity-weighted median survival age was 64.6 years on rapamycin vs 64.3 years without it (95% CI for difference -1.6, 2.0 years) over a median 11 years. Caveats: daily immunosuppressive dosing, not intermittent geroprotective dosing, and residual confounding by indication.

Whole-blood NAD levels track cardiovascular disease but not chronological age

Nat Metab · Cross-sectional cohort, 1,068 adults · NAD as an aging biomarker

Across 1,340 measurements, whole-blood NAD did not decline with age in healthy people; it was stable in males and rose in females. It was reduced in cardiovascular disease and hypertension. This undercuts the age-related NAD decline rationale for supplementation, but the data are associational and say nothing about treatment benefit.

Altered protein cysteine reactivity in senescent cells is a vulnerability for senolysis to ameliorate aging

Nat Aging · Preclinical mouse and cell study · Senolytics

Senescent cells accumulate glutathione disulfide through glutathione reductase downregulation, which leaves them selectively vulnerable to vanadyl sulfate, an oral mineral supplement that drives protein misfolding and cell death. Oral dosing improved several age-related phenotypes and extended lifespan in male mice. The evidence is mouse-only, in one sex, and there are no human safety data at these doses.

General Medicine

Three themes ran through general medicine this week. First, incretin-based obesity drugs keep moving into new indications and better ways of using them. Retatrutide reduced knee osteoarthritis pain alongside roughly 20% weight loss in two phase 3 trials. A trial pairing tirzepatide with resistance training showed that exercise preserves lean mass and strength without blunting weight loss, a practical answer to the sarcopenia concern. Second, two large trials questioned whether doing more helps. Prepare for Kidney Care found conservative kidney management in frail older patients delivered quality-adjusted survival statistically similar to preparing for dialysis. OPTIMAL-DIABETES found that a systolic target below 120 mm Hg in type 2 diabetes did not significantly reduce cardiovascular events, unlike SPRINT in nondiabetic patients. Third, a JAMA cohort showed that refusal of newborn vitamin K prophylaxis has roughly doubled in the US, with more than twice the risk of intracranial bleeding in infants who did not receive it.

Preparing for responsive management versus renal dialysis in older people living with kidney failure and frailty or multimorbidity in the UK (Prepare for Kidney Care): a two-arm, superiority, parallel-group, open-label, pragmatic, phase 3, individually randomised controlled trial

Lancet · Pragmatic open-label RCT, 445 patients · Kidney failure in frail older adults

UK trial in older frail or multimorbid patients with eGFR below 15: conservative responsive management vs dialysis preparation gave similar QALYs (adjusted difference -0·147, 95% CI -0·350 to 0·056).

Intensive Blood-Pressure Lowering in Patients with Type 2 Diabetes

N Engl J Med · Open-label RCT, 9476 patients · Blood pressure targets in type 2 diabetes

Brazilian OPTIMAL-DIABETES trial: a systolic target below 120 vs below 140 mm Hg did not significantly reduce major cardiovascular events over a median 4.0 years (HR, 0.90; 95% CI, 0.77 to 1.06).

Retatrutide Once Weekly in Adults with Knee Osteoarthritis Pain and Obesity

N Engl J Med · Two phase 3 RCTs, 1019 patients · Knee osteoarthritis and obesity

In TRIUMPH-4, retatrutide cut WOMAC pain vs placebo (difference -1.83 at 9 mg) with -23.7% weight loss at 12 mg; serious adverse events were more common.

Resistance exercise to attenuate lean mass loss during tirzepatide therapy for obesity: a randomized controlled trial

Nat Med · RCT, 120 patients · Obesity pharmacotherapy and lean mass

Adding thrice-weekly resistance exercise to tirzepatide preserved lean mass (between-group difference 1.09 kg) and increased strength. Weight loss was unchanged at 40 weeks, but the trial was small and single-country.

Nonreceipt of Newborn Vitamin K and Risk of Bleeding

JAMA · Retrospective cohort, ~5.2 million infants · Newborn vitamin K prophylaxis

In US Epic Cosmos data, vitamin K nonreceipt rose from 3.1% to 6.8% between 2017 and 2025. Unprophylaxed infants had higher intracranial bleeding risk (ARR, 2.43; 95% CI, 2.26-2.62).

Verification Notes

- 498 candidate papers screened from PubMed; 21 selected (9 with a full write-up).
- Every selected PMID, title, author list and journal is taken directly from PubMed. Design and Results lines are drawn from the abstract and their numbers checked against it automatically. Context lines reflect the model's knowledge of the prior literature and are not machine-verified.
- AI in Medicine, PMID 42995231: the write-up cites 3, not found verbatim in the PubMed abstract. Check against the paper.

Selected and written by claude-opus-5-5 from PubMed records added Nov 04 – Nov 11, 2026.